ASSIGNMENT NO: 4

Title:Write a program in C++ to perform following operations on complex numbers Add, Subtract,Multiply. Design the class for complex number representation and the operations to be performed. The objective of this assignment is to learn the concepts classes and objects.

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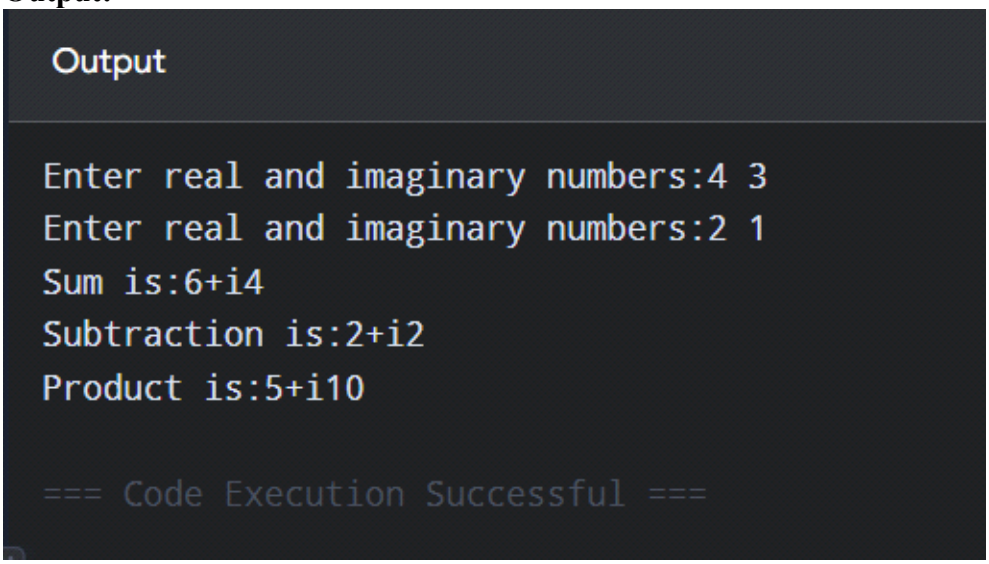
Program:

```
#include<iostream>
#include<math.h>
using namespace std;
class complex
{
   private:
   int real,img;
   public:
   void read();
   void display();
   complex multiply(complex);
   void add(complex,complex);
    void subtract(complex,complex);
};
void complex::read()
```

```
{
   cout<<"Enter real and imaginary numbers:";
   cin>>real>>img;
}
void complex::display()
{
   if(img>0)
   {
      cout<<real<<"+i"<<img;
   }
   else
   {
      cout<<real<<"-i"<<img;
   }
}
void complex::add(complex C1,complex C2)
{
  real=C1.real+C2.real;
  img=C1.img+C2.img;
}
void complex::subtract(complex C1,complex C2)
{
  real=C1.real-C2.real;
  img=C1.img-C2.img;
```

```
}
complex complex::multiply(complex C1)
{
    complex product;
    product.real=(real*C1.real)-(img*C1.img);
    product.img=(real*C1.img)+(img*C1.real);
    return product;
}
int main()
{
    complex C1,C2,C3,C4,C5;
    C1.read();
    C2.read();
    C3.add(C1,C2);
    cout<<"Sum is:";
    C3.display();
    C4.subtract(C1,C2);
    cout<<"\nSubtraction is:";
    C4.display();
    C5 = C1.multiply(C2);
    cout<<"\nProduct  is:";
    C5.display();
    return 0;
}
```

Output:


Output
Enter real and imaginary numbers:4 3
Enter real and imaginary numbers:2 1
Sum is:6+i4
Subtraction is:2+i2
Product is:5+i10
=== Code Execution Successful ===